Midterm

7:00-9:00pm, 12 March 2026

Your First Name: Adithya **Your Last Name:** Giri

SIGN Your Name: Adithya **Your SID Number:** 3041015344

Your Exam Room: VLSB 2050

Name of Person Sitting on Your Left: N/A

Name of Person Sitting on Your Right: Brandon Qi

Name of Person Sitting in Front of You: N/A

Name of Person Sitting Behind You: Ethan Thomas

Instructions:

(a) *As soon as the exam starts,* **please write your student ID in the space provided at the top of every page!** *(We will remove the staple when scanning your exam.)*

(b) *There are 13 pages (2-sided) on the exam. Notify a proctor immediately if a page is missing.*

(c) *We will* **not** *grade anything outside of the space provided for a question (i.e., either a designated box if it is provided, or otherwise the white space immediately below the question).* **Be sure to write your full answer in the box or space provided!** *Scratch paper is provided on request; however, please bear in mind that nothing you write on scratch paper will be graded!*

(d) **The questions vary in difficulty, so if you get stuck on any question you are strongly advised to leave it and return to it later.** *In particular, you may find some of the later questions easier than some of the earlier ones. Point values are not an indication of difficulty. There is no penalty for incorrect answers.*

(e) *In the interests of fairness, proctors will* **not** *answer clarifying questions during the exam. If you believe there is an ambiguity or error in a problem, write down what you think is the most appropriate response and move on. Any legitimate such issues will be handled during grading.*

(f) *You may consult* **one two-sided handwritten "cheat sheet"** *of notes. Apart from that, you may not look at any other materials. Calculators, phones, computers, and other electronic devices are* **not** *permitted.*

(g) *You may use, without proof, theorems and lemmas that were proved in the notes and/or in lecture.*

(h) *You have 120 minutes: there are 13 questions on this exam worth a total of 102 points.*

[exam starts on next page]

1. Feline True/False [4 points].

For this problem, assume that:

- If a cat has green eyes then it has a bushy tail
- A cat isn't friendly unless it has green eyes
- Not all cats have bushy tails

GE → BT ∀cat (! BushyTail)
GE → FR

Based on this information, say whether each of the following statements is definitely True, definitely False, or Undetermined by the given information.

Answer by shading the correct bubble; no justification; no penalty for incorrect answers.

TRUE	FALSE	UNDET		
●	○	○	Not all cats are friendly	*2pts*
○	○	●	There exists a cat with green eyes	*2pts*

2. Logic True/False [10 points].

In this question you should assume that P and Q are propositions or predicates with the appropriate parameters. The variables n, m are assumed to be quantified over the universe of natural numbers. For each implication or equivalence, say whether it is True or False.

Answer by shading the correct bubble; no justification; no penalty for incorrect answers.

TRUE	FALSE		
●	○	$((P \vee \neg Q) \wedge (\neg P \vee Q)) \equiv (P \Leftrightarrow Q)$	*2pts*
○	●	$\neg(P \wedge \neg Q) \equiv (Q \Rightarrow P)$	*2pts*
○	●	$(P(0) \wedge (\forall n)(P(n) \Rightarrow P(n+2)) \Rightarrow \forall n P(n)$	*2pts*
●	○	$(P(0) \wedge (\forall n > 0)(P(n) \Rightarrow P(n-1)) \wedge (\forall n)(P(n) \Rightarrow P(n+2))) \Rightarrow \forall n P(n)$	*2pts*
○	●	$(\forall m)(\exists n)P(n,m) \wedge (\forall n)(\exists m)P(n,m) \Rightarrow \forall m \forall n P(n,m)$	*2pts*

[exam continues on next page]

3. Induction [8 points].

Anna writes down an arbitrary positive integer n on a blackboard. She then repeatedly performs the following operation, where k is the number currently on the board: *8pts*

- if k is a multiple of 3 then replace k by $k/3$; else replace k by $k+1$

Prove by induction that, for any starting integer n, Anna will always eventually write the number 1 on the board. [**Hint:** Use strong induction. Consider three cases according to whether $n \equiv 0, 1$ or $2 \pmod 3$.]

Base Case: $n=1$, Anna will write 1

Base Case: $n=2$, Anna will add 1 to 2 to get 3, and then divide 3 by 3 to get 1.

Inductive Case:

$n \equiv 0 \pmod 3$: Anna divides by 3, so the next written number is $n/3$, $n/3 < n$. She will eventually write 1 in this case by strong induction.

$n \equiv 1 \pmod 3$: Anna will add 1 until she reaches $n+2$, at which point $n+2$ is divisible by 3, she will divide by 3. $\frac{n+2}{3} < n$, $n > 2$, hence she will eventually write 1 by strong induction.

$n \equiv 2 \pmod 3$: Anna will add 1, at which point $n+1$ is divisible by 3. She will then divide by 3. $\frac{n+1}{3} < n$, $n > 2$, hence by strong induction Anna will eventually write 1.

[exam continues on next page]

4. Algebraic Numbers T/F [10 points].

We call a real number *algebraic* if it is a root of a non-zero polynomial with *integer* coefficients. Answer the following questions by shading the correct bubble; no explanations needed; no penalty for incorrect answers.

TRUE	FALSE		
●	○	all rational numbers are algebraic	*2pts*
○	●	all algebraic numbers are rational	*2pts*
●	○	$\sqrt{2}$ is algebraic	*2pts*
○	●	the set of algebraic numbers is countable	*2pts*
●	○	if $x \neq 0$ is algebraic, then $\frac{1}{x}$ is algebraic	*2pts*

[exam continues on next page]

5. Stable Matching Miscellanea [12 points].

(a) Consider the following preference lists for four jobs, A, B, C, D, and four candidates, 1, 2, 3, 4:

Job	Candidates			
A	1	2	3	4
B	2	3	1	4
C	3	4	1	2
D	1	4	2	3

Candidate	Jobs			
1	B	D	A	C
2	C	A	B	D
3	A	B	D	C
4	D	C	A	B

What are the optimal and pessimal candidates for job B? Write your answers in the boxes below; no justification required. *4pts*

Optimal candidate for B: 2

Pessimal candidate for B: 1

(b) Suppose a pool of n candidates contains k candidates from UC Berkeley, and all k of those candidates are placed (in possibly different orders) in the top k positions on the preference lists of all n jobs. Prove that, in *any* stable matching, every candidate from Berkeley must be matched with one of their top k jobs. *4pts*

Proof by Contradiction:

Assume a candidate c from Berkeley is not matched with one of their top k jobs. Then there exists a candidate not from Berkeley, n, paired with a job j the top k of c's job preference list. Since the top k of every company's preference list comprises of only Berkeley students, j must prefer c over n. c also prefers j over its current job, since j is in c's top k jobs. Hence, (c, j) is a rogue couple. □

(c) Suppose job J is rejected by candidate C in the propose-and-reject algorithm with jobs proposing. Prove that J is never paired with C in *any* stable matching. *4pts*

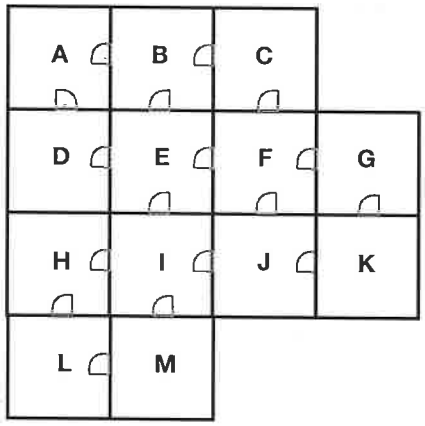
Direct Proof:

Propose-and-reject with jobs proposing provides a candidate-pessimal stable matching. If job J is rejected by candidate C, they like their job in hand, J', more than J. The job they end up with will be atleast as good as J'. Since the candidate likes the job they ended up with in a candidate-pessimal matching more than J, there is no stable matching where J and C are paired together. □

[exam continues on next page]

6. Euler's Dollhouse [5 points].

The diagram below show's the floor plan of the upper level of a dollhouse. Rooms are labeled with upper case letters and red symbols indicate doors between rooms; doors can be traversed in both directions.



(a) Euler wants to start in room A and walk through all the doors exactly once (one direction only for each door), ending up back in room A. He soon discovers that this isn't possible, but if he adds a secret passageway between two rooms that are not adjacent to each other (where the passageway is treated like a door between those rooms) it does become possible. Between which two rooms should Euler add the passageway? Write your answer in the box below; no justification required. *2pts*

B, J

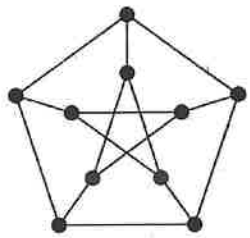
(b) Euler adds the secret passageway from part (a), but then decides that the house has too many doors and wants to block up some of them to create more wallspace. He wants to keep the property in part (a), i.e., that it's possible to walk through all the doors, and he doesn't want to isolate any rooms or block off the passageway. Which set of **four** doors can Euler block off to achieve this? Indicate your answer by checking **exactly four** squares in the list of doors below. No justification required; zero points if you check more than four squares. *3pts*

☐ A-B	☐ A-D	☐ B-C	☐ B-E
☐ C-F	☐ D-E	■ E-I	■ E-F
☐ F-G	■ F-J	☐ G-K	☐ H-I
☐ H-L	■ I-J	☐ I-M	☐ J-K
☐ L-M			

[exam continues on next page]

7. The Petersen Graph [8 points]

Consider the following graph with 10 vertices and 15 edges, known as the Petersen graph.



In this problem you will prove that the Petersen graph is not planar.

(a) Recall from class that if a graph is planar then it must satisfy $e \leq 3v - 6$, where e and v are the numbers of edges and vertices, respectively. Show that this fact is **not** enough to prove that the Petersen graph is not planar. *1pt*

$e = 15 \quad v = 10$

$15 \leq 3(10) - 6 \leq 24$ ✓ $\qquad 15 \leq 24$ ✓

(b) Suppose that the length of a shortest cycle in a graph G is 5. Prove that any planar drawing of G must satisfy $2e \geq 5f$, where e is the number of edges and f is the number of faces. [**Hint:** Recall from class a similar argument for bipartite graphs, which have shortest cycle length at least 4.] *4pts*

The number of sides per face is atleast the length of the shortest cycle, so each face has atleast 5 sides. Summing up the sides of all faces double counts each edge, so $2e \geq 5f$.

(c) Use part (b) to derive a stronger inequality than the one in part (a) for planar graphs with shortest cycle length at least 5. Then use your condition to prove that the Petersen graph is not planar. [**Note:** The length of a shortest cycle in the Petersen graph is 5.] *3pts*

$2e \geq 5f$, $v + f = e + 2$, $f = e - v + 2$, $5f = 5e - 5v + 10$

$2e \geq 5e - 5v + 10$

$-3e \geq -5v + 10$ $\qquad 45 \leq 40 \Rightarrow$ not true

$3e \leq 5v - 10$ $\qquad$ Hence, the petersen graph is not planar.

[exam continues on next page]

8. Modular Arithmetic and RSA–Short Answers [10 points].

Write your answer only in the box provided (no justification). When working in arithmetic mod m, your answer should be an integer in the range $\{0, 1, \ldots, m-1\}$.

(a) Solve the equation $3x \equiv 5 \pmod{16}$. *2pts*

$\boxed{7}$

(b) Calculate $2^{39} \pmod{37}$. *2pts*

$2^{36} \equiv 1 \pmod{37}$

2^3

$\boxed{8}$

(c) Calculate $2025^{503} \pmod{2026}$. *2pts*

$\boxed{2025}$

(d) Alice is sending a message to Bob using RSA. If Bob's public key is $e = 3$; $N = 55$ and Alice's message is 10, what encrypted message should Alice send? *2pts*

$\boxed{10}$

(e) In the same scenario as part (d), what is the value of the decryption key d that Bob should use to decrypt the message? *2pts*

$\boxed{27}$

[exam continues on next page]

9. Intensive Swim Workout [8 points].

A swimmer swims in a pool where 70 laps is equivalent to 1 mile. (By "lap" here we mean one length of the pool.) She uses the following exhausting workout regime:

- In alternate laps, she puts her fins on and takes them off.
- She cycles through five breathing patterns (labeled 1,2,3,4,5), switching to the next one at the end of each lap.
- She swims seven different strokes (1=freestyle, 2=breaststroke, 3=butterfly, 4=backstroke, 5=front kick, 6=back kick, 7=sidestroke), switching to the next one at the end of each lap.

She always starts her workout with fins off, breathing pattern 1, freestyle.

One day she loses count of how many laps she swam, but she does know she swam more than one mile and less than two miles. She also knows that her last lap was with fins on, breathing pattern 3, and back kick.

(a) Explain clearly why she has enough information to deduce exactly how many laps she swam. *5pts*

Assume the number of laps she swam is l. If she has her fins on, $l \equiv 1 \pmod 2$, If she is in breathing pattern 3, $l \equiv 3 \pmod 5$, if she is on back kick, $l \equiv 6 \pmod 7$. Since 2, 5, and 7 are relatively prime, there exist an $l \pmod{70}$, where $l \equiv 1 \pmod 2$, $l \equiv 3 \pmod 5$, and $l \equiv 6 \pmod 7$ by the chinese remainder theorem. Since she swam between 1 and 2 miles, and $l' \equiv l \pmod{70}$ exists, $l' + 70 = l$, so she can figure out how many laps she swam.

(b) How many laps did she swim? Write your final answer in the box; no working required. [**Note:** You may find it easier to do this calculation by hand rather than using a standard algorithm.] *3pts*

Number of laps: $\boxed{83}$

[exam continues on next page]

10. Leaking Secrets [6 points].

Consider the secret sharing scenario in class, where a Commander wants to share a **2-digit** secret code with n generals in such a way that any $k = 3$ generals can recover the code, but 2 generals cannot. The Commander doesn't like modular arithmetic, and prefers to work over the integers. Thus he selects a polynomial $p(x)$ of degree 2 of the form

$$p(x) = ax^2 + bx + s,$$

where s is the secret code and a, b are integers, and gives each general a point on this polynomial. We shall see in this problem that the Commander's scheme may actually reveal the secret, even to two generals!

(a) Suppose that two of the generals have the points (5,25) and (15,15). Write down two equations for the coefficients a, b, s, and solve for s as a function of a. *4pts*

$$\boxed{\begin{aligned} 25 &= 25a + 5b + s \\ 15 &= 225a + 15b + s \end{aligned}} \quad 3$$

$75 = 75a + 15b + 3s$

$15 = 225a + 15b + s$

$60 = -150a + 2s$

$150a + 60 = 2s$

$\boxed{s = 75a + 30}$

Use part (a) to deduce the value of the secret s. [Recall that s is a 2-digit integer.] Write your answer in the box; no justification required. *2pts*

Value of secret s: $\boxed{30}$

[exam continues on next page]

11. Error Correction [8 points].

Suppose we want to send a message of n packets (viewed as an integer modulo some large prime q) across an unreliable link. We anticipate that up to 7 packets may be corrupted, so we send a codeword consisting of $n+m$ points $P(1), \ldots, P(n+m)$ on some $GF(q)$ polynomial $P(x)$ of degree $n-1$. Answer the following questions about this scenario.

(a) What value should we choose for m? Write your answer in the box; no justification required. *1pt*

$\boxed{14}$

(b) Suppose that in reality 10 packets are corrupted, and we receive the values $r_1, r_2, \ldots, r_{n+m}$, where m is as in part (a). It is still possible to recover the polynomial P if we are told the locations of some of the errors. What is the minimum number of error locations we need to know to achieve this? Write your answer in the box; no justification. *3pts*

$\boxed{6}$

(c) Let ℓ be the minimum number of error locations you found in part (b), and suppose we are told that those error locations are at positions $1, 2, \ldots, \ell$. Briefly explain how you would recover the original message given this information. To receive full credit, be sure to specify the degrees of the polynomials $E(x)$ and $Q(x)$ and the points on which they should be evaluated. Your answer should be in terms of the variables m and ℓ defined in parts (a) and (b), respectively. You may assume familiarity with the Berlekamp-Welch algorithm, but you should indicate how to apply it in this setting. *4pts*

Remove all the 6 error locations. This leaves you with $n+14-6=n+8$ points and 4 corrupted points at unknown locations. Construct a polynomial $Q(x)$ with degree $n+7$, where $Q(x)=E(x)P(x)$, where $P(x)$ is our original polynomial and $E(x)=(x-e_1)(x-e_2)(x-e_3)(x-e_4)$, where e_1 through e_4 are the unknown error locations. Compute coefficients for $Q(x)$ using $n+8$ points, and figure out $E(x)$ from the equations $Q(i)=r_iE(i)$ computed at 4 of the $n+8$ points. Once you find 4 points for $E(x)$, you can figure out the coefficients of $E(x)$ using Lagrange interpolation. Then $P(x)=\frac{Q(x)}{E(x)}$, recovering your original message.

[exam continues on next page]

12. Countability [10 points].

For each of the following sets, determine if the set is finite, countably infinite, or uncountably infinite. Answer by shading the correct bubble; no explanations needed; no penalty for incorrect answers.

FINITE	CTBLY INF	UNCTBLE		
○	●	○	The set of solutions $x \in \mathbb{Z}$ to the equation $3x \equiv 1 \pmod 9$.	*2pts*
○	○	●	The set of functions from $\mathbb{R}$ to $\mathbb{N}$.	*2pts*
○	●	○	The set of numbers $y \in \mathbb{R}$ such that y^4 is an integer.	*2pts*
●	○	○	The set of (simple) graphs with 100 vertices.	*2pts*
○	●	○	The set of Python programs that halt on every input.	*2pts*

[exam continues on next page]

13. Computability [3 points].

Consider the following problem, which we call the equivalence problem, EQUIV:

Input: Two programs, P_1, P_2

Question: On every input x, do P_1 and P_2 either both Halt on x or both Loop on x?

We can show that EQUIV is undecidable by reducing the Halting Problem, HALT, to it as follows.

program TestHalt(P, x)
  construct program P_1 which, on input y, simulates P on x
  construct program P_2 which, on input y, ... $(*)$ .....
  return TestEquiv(P_1, P_2)

Which command should be inserted at position $(*)$ in the above code to make this a valid reduction? Write your answer in the box provided; no justification. *3pts*

always halts

[End of exam!]